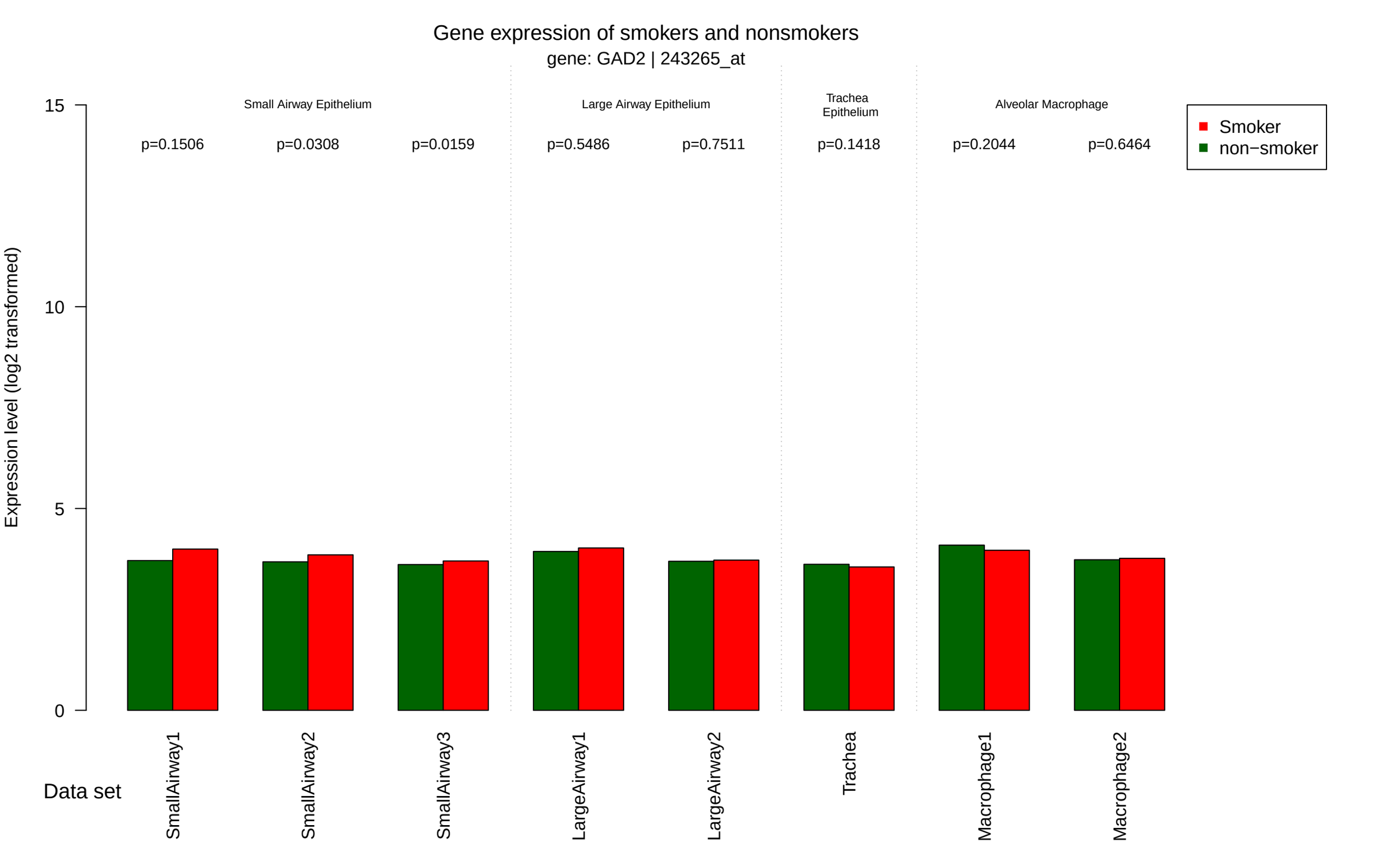
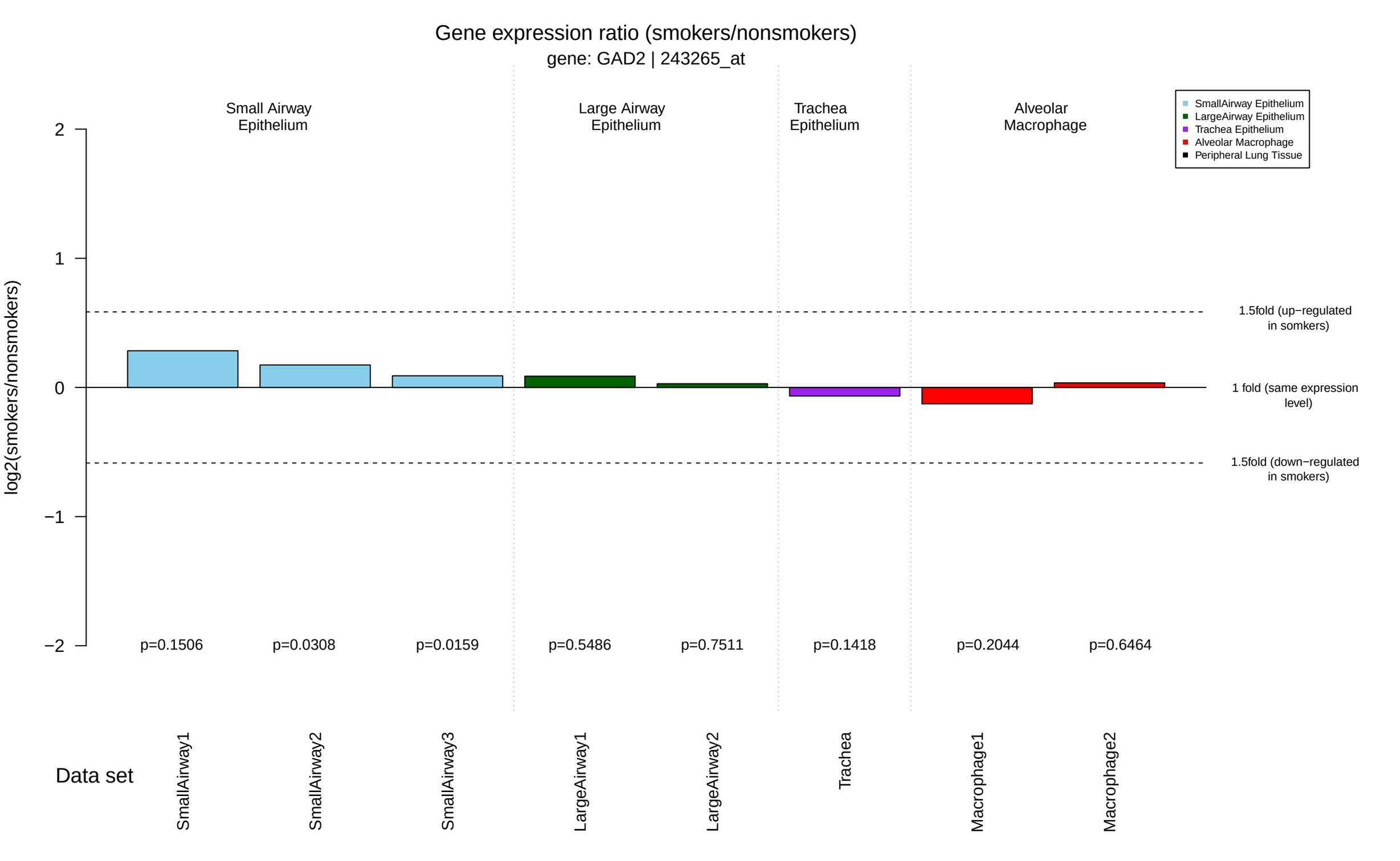
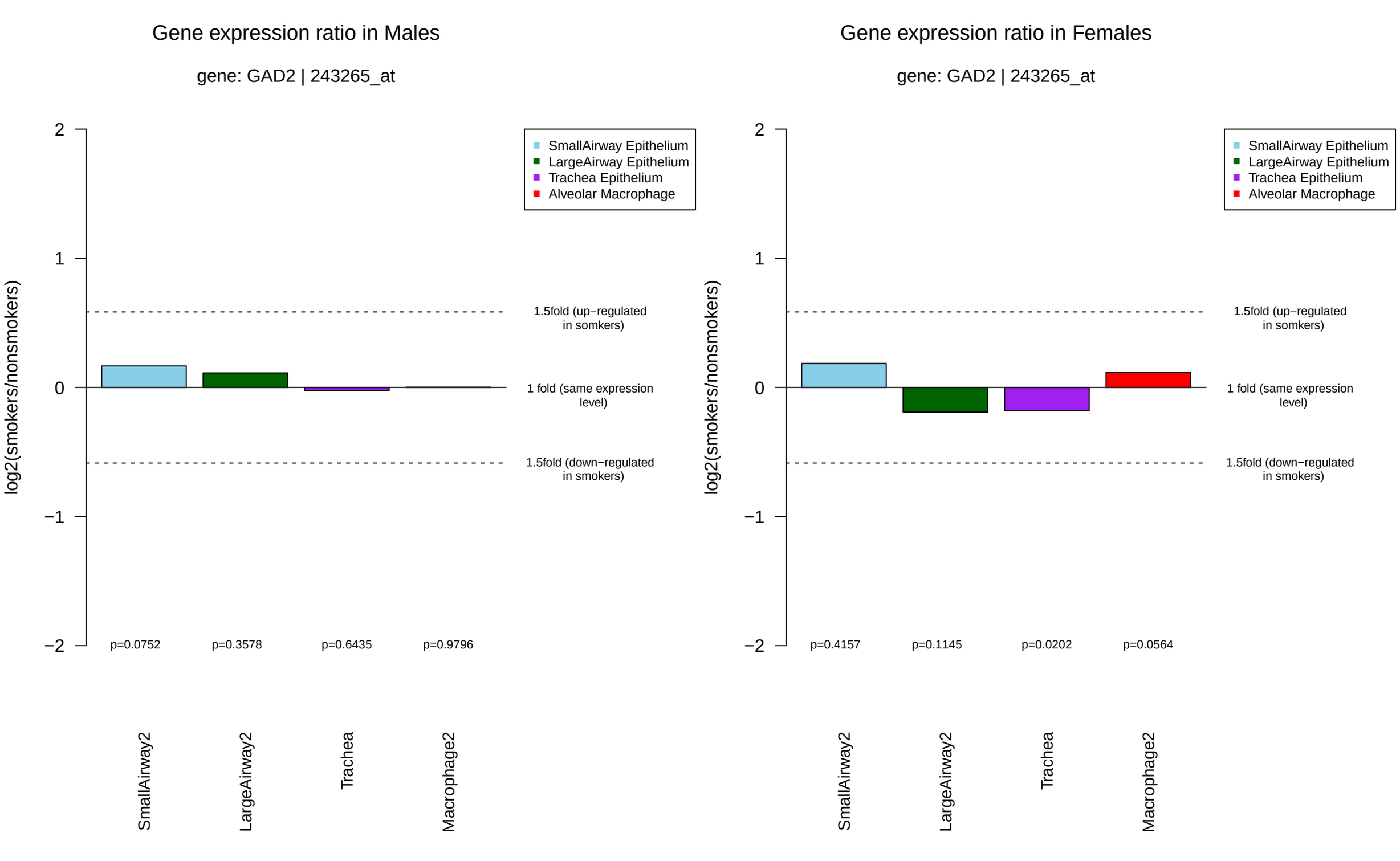
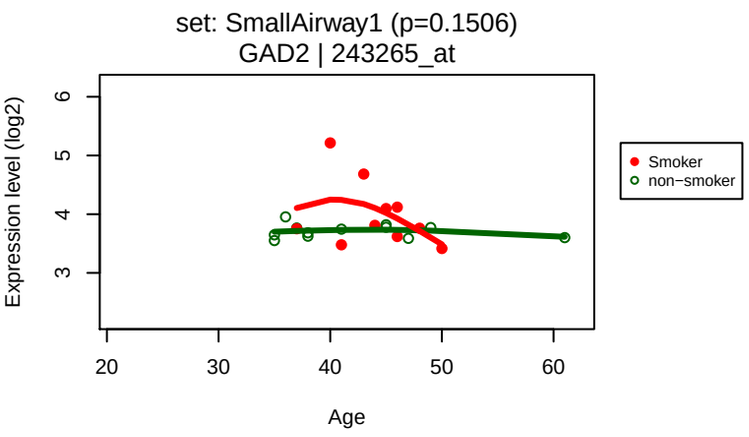
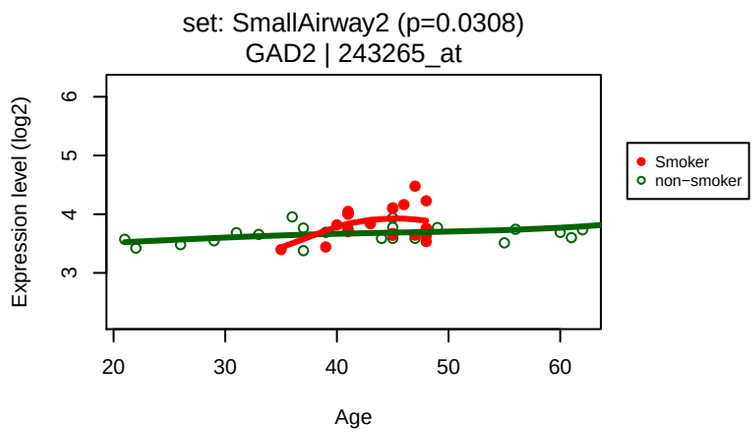
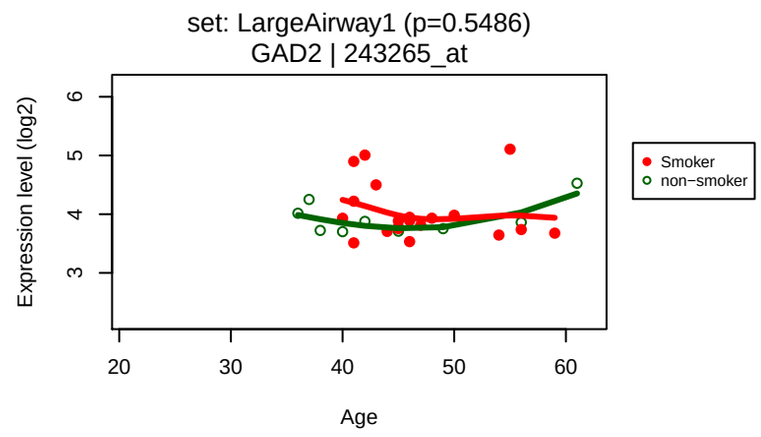

Gene expression of smokers and nonsmokers
gene: GAD2 | 243265_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1506
p=0.0308
p=0.0159
p=0.5486
p=0.7511
p=0.1418
p=0.2044
p=0.6464
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

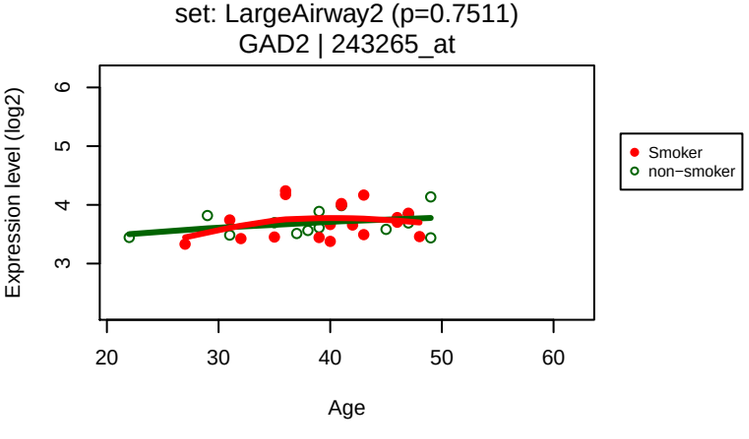

Gene expression ratio (smokers/nonsmokers)
gene: GAD2 | 243265_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1506
p=0.0308
p=0.0159
p=0.5486
p=0.7511
p=0.1418
p=0.2044
p=0.6464
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

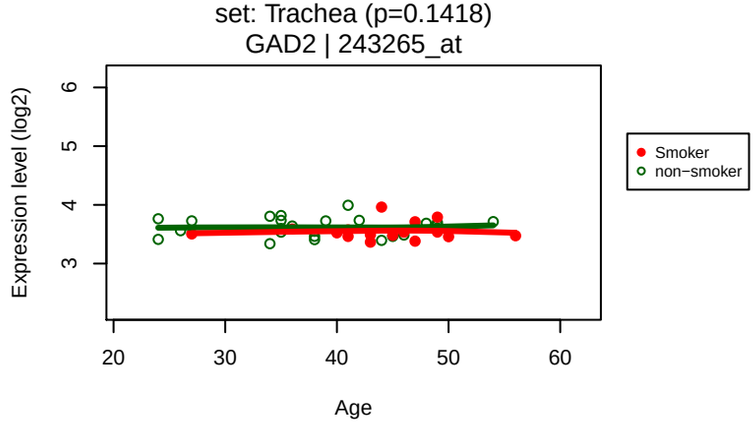

Gene expression ratio in Males
gene: GAD2 | 243265_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0752
p=0.3578
p=0.6435
p=0.9796
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAD2 | 243265_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4157
p=0.1145
p=0.0202
p=0.0564
SmallAirway2
LargeAirway2
Trachea
Macrophage2

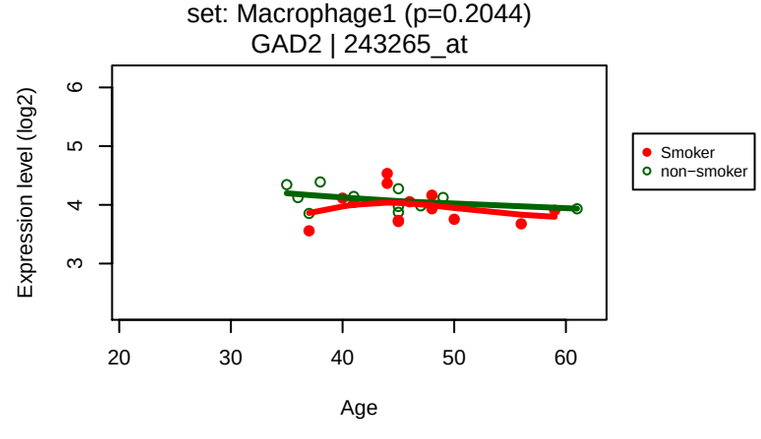

set: SmallAirway1 (p=0.1506)
GAD2 | 243265_at
Expression level (log2)
Age
Smoker
non−smoker

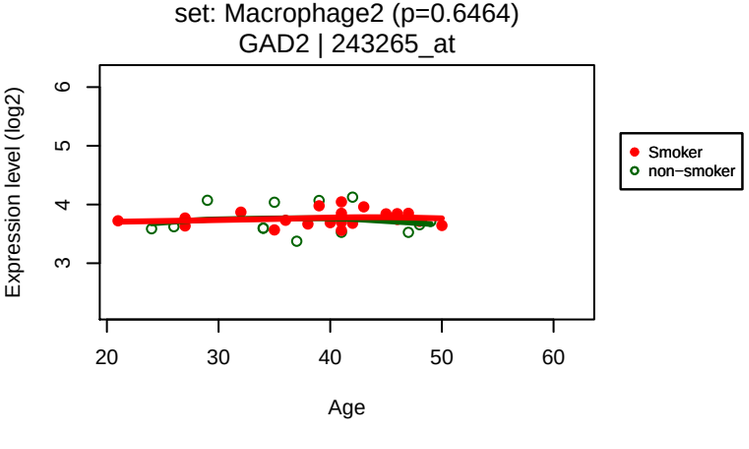

set: SmallAirway2 (p=0.0308)
GAD2 | 243265_at
Expression level (log2)
Age
Smoker
non−smoker

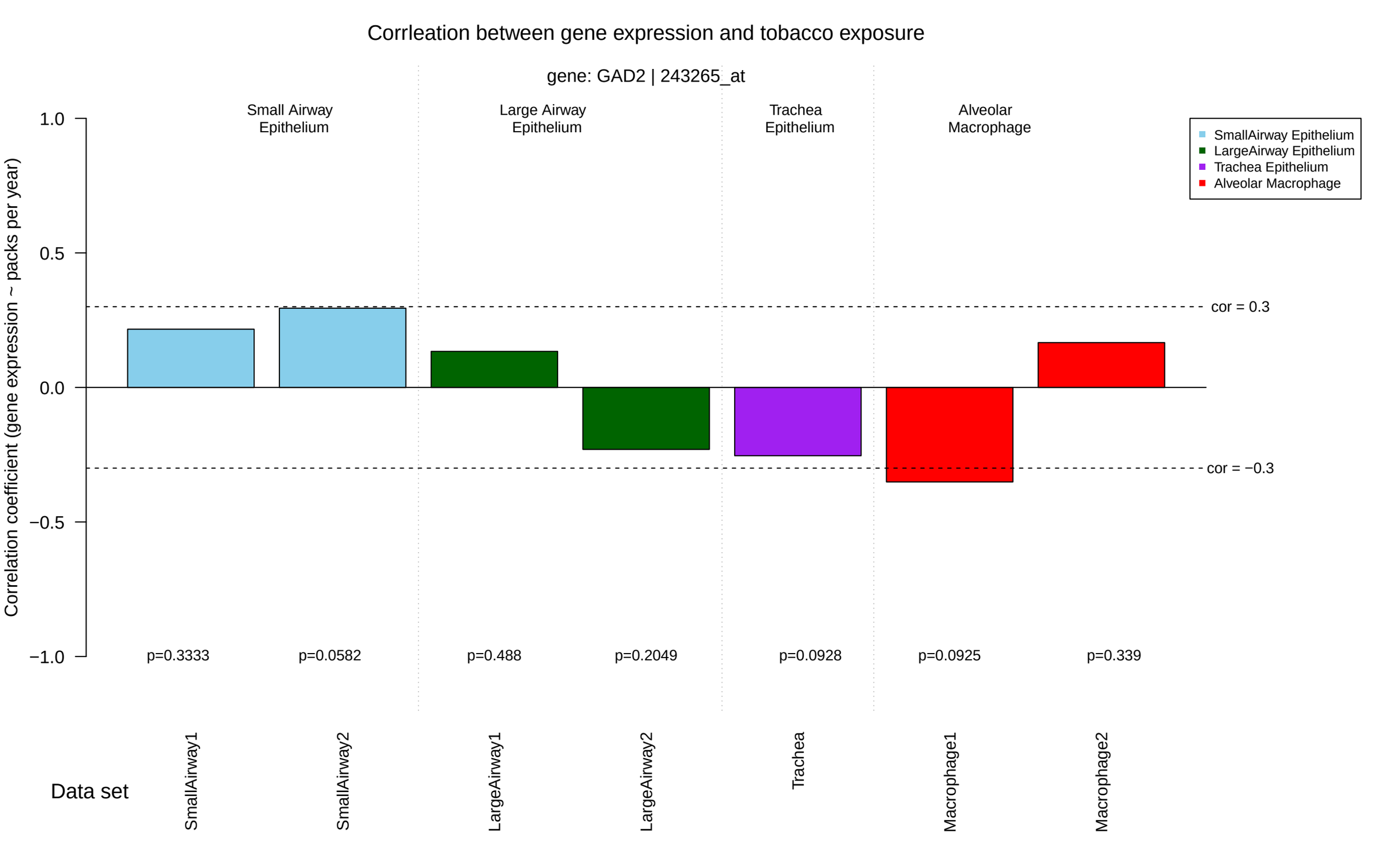

set: LargeAirway1 (p=0.5486)
GAD2 | 243265_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7511)
GAD2 | 243265_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1418)
GAD2 | 243265_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2044)
GAD2 | 243265_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6464)
GAD2 | 243265_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAD2 | 243265_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3333
p=0.0582
p=0.488
p=0.2049
p=0.0928
p=0.0925
p=0.339
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2